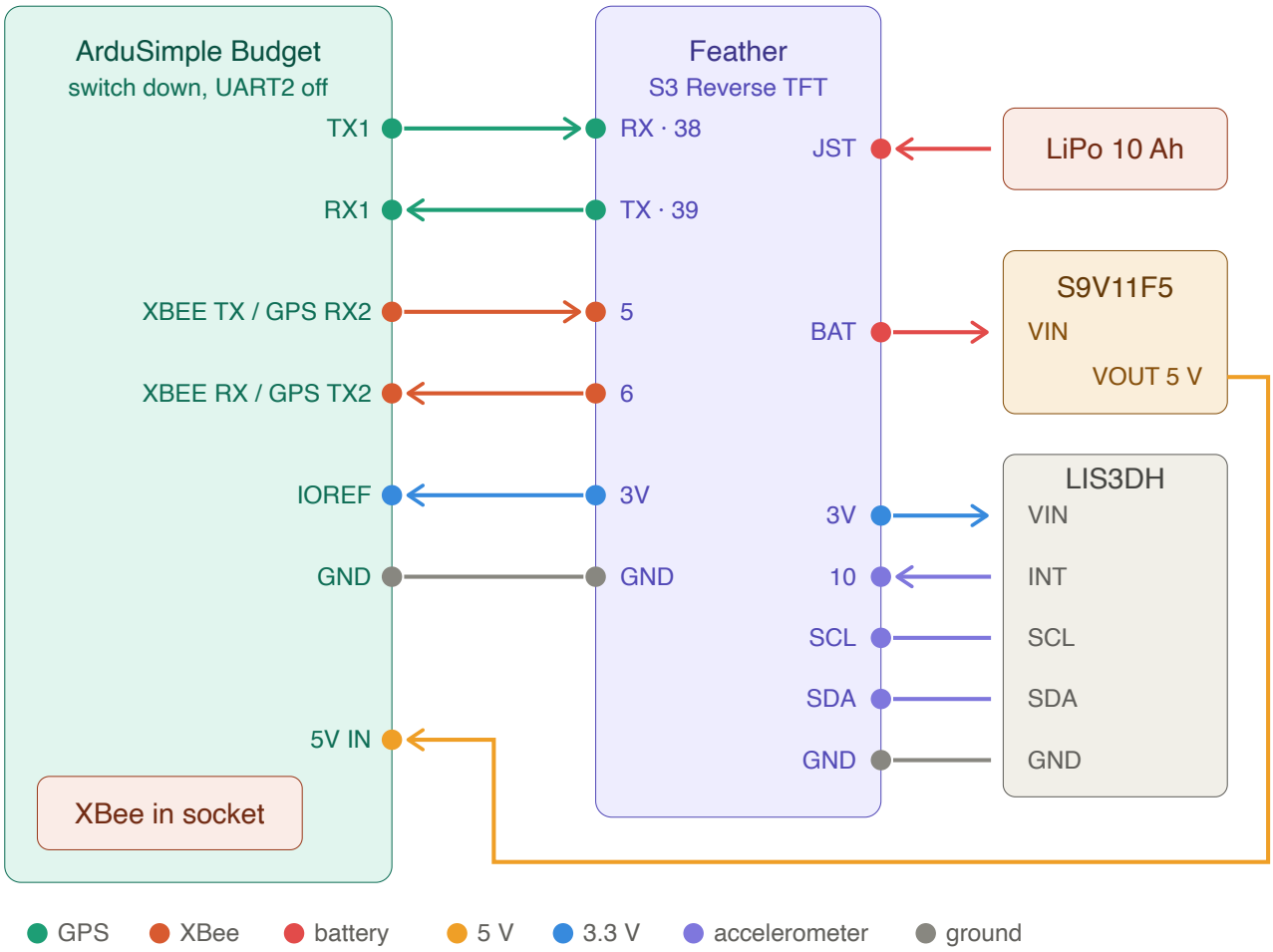
ESP32 rover wiring

Feather ESP32-S3 Reverse TFT (5691), ArduSimple simpleRTK2B Budget with the XBee in its socket, LIS3DH, Pololu S9V11F5, 10,000 mAh LiPo. Firmware default pins.

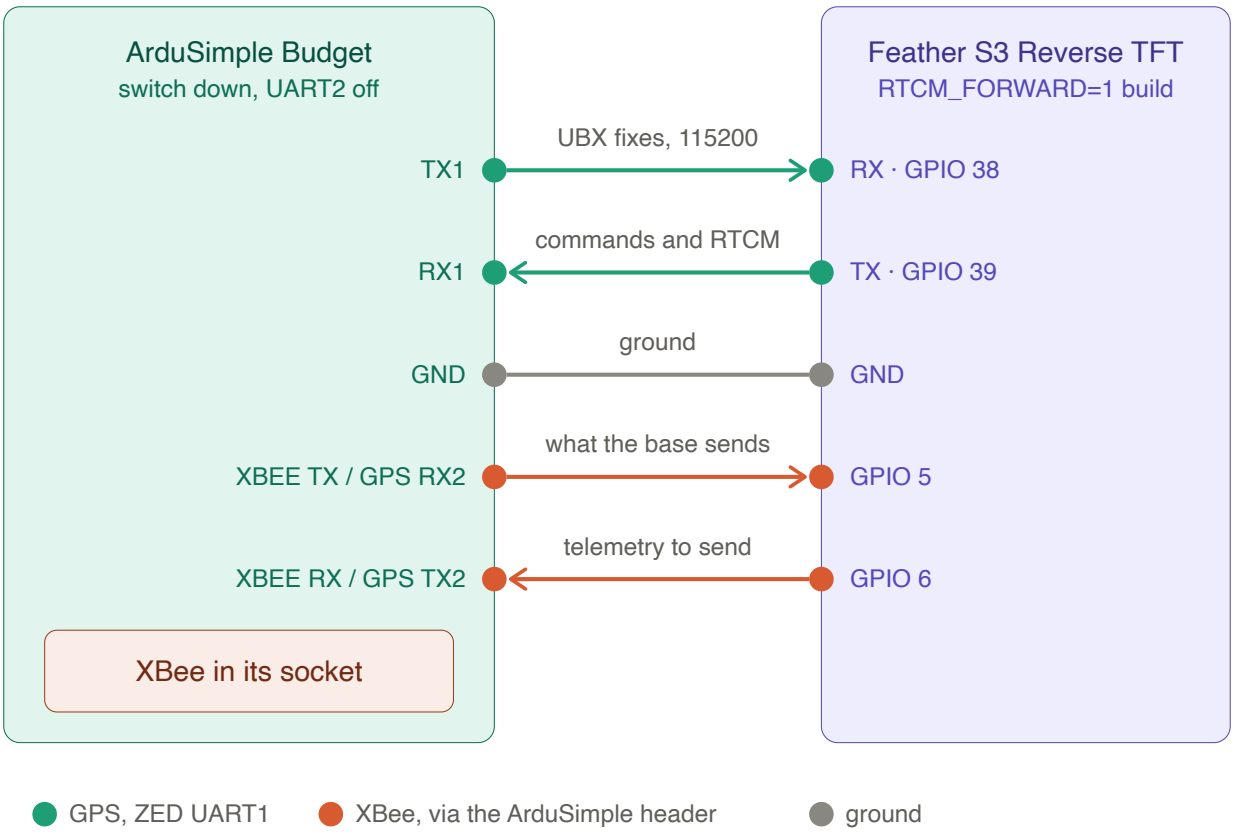
All connections



All grounds common (Feather, ArduSimple, S9V11F5, LIS3DH). S9V11F5 EN left open.

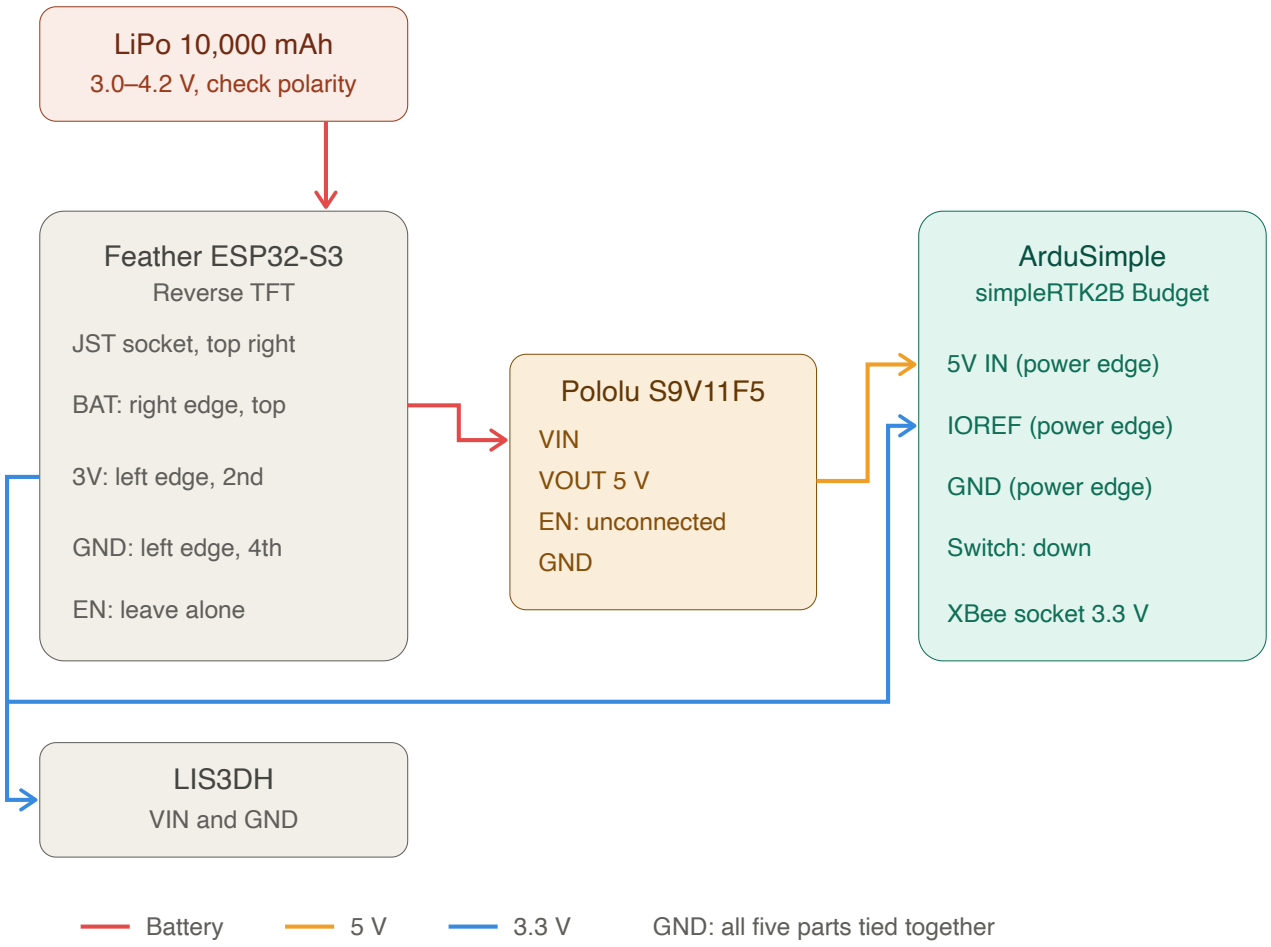
Every wire on one page. Pin positions on the Feather's edges are on page 4; physically, both 3V pads and GND are on the left edge (2nd, 3rd and 4th from the top), so the LIS3DH's VIN and GND wires go to the left edge, and BAT is at the top of the right edge.

1. Signal wiring



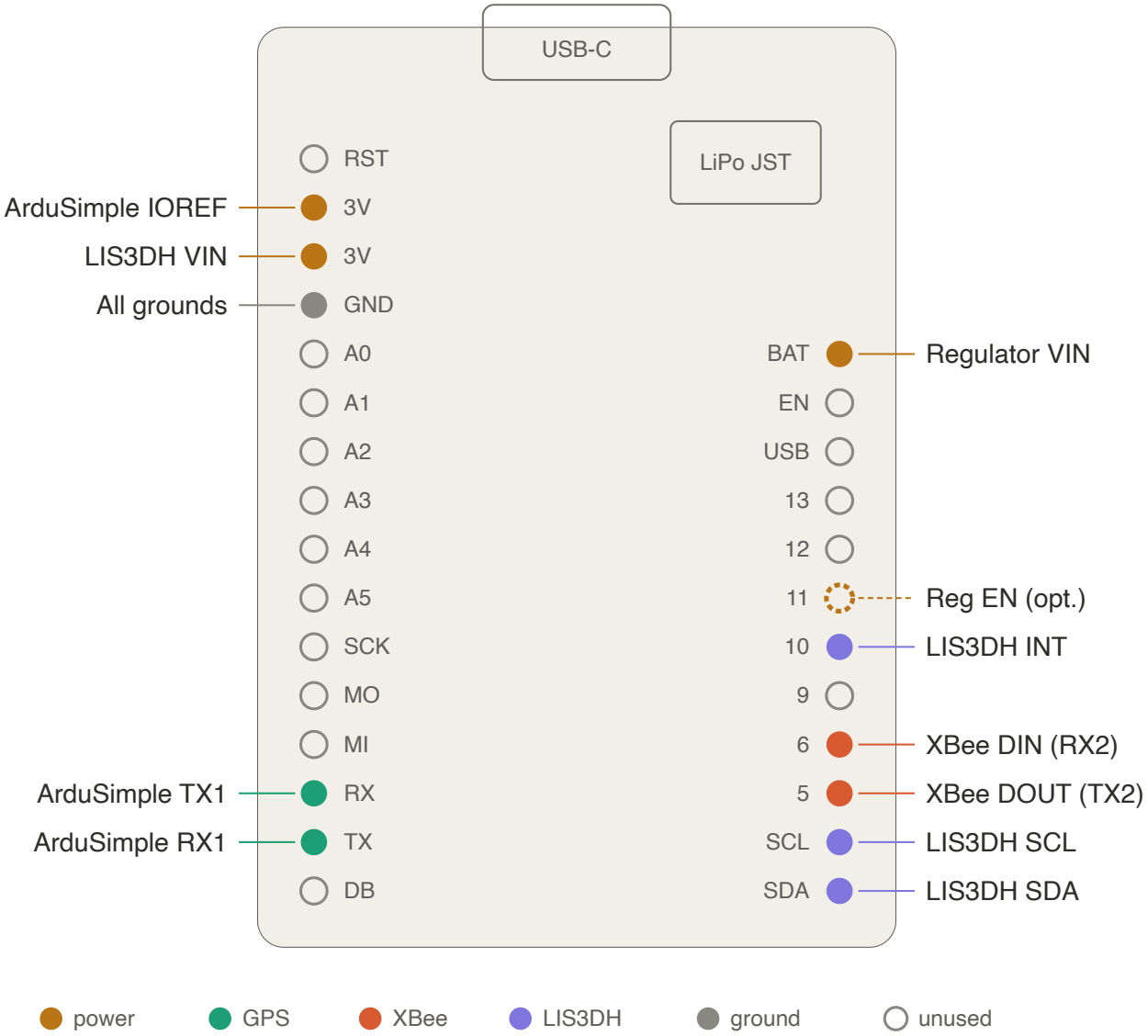
The XBee stays in the ArduSimple's socket. The ZED's UART2 is switched off (`ubx_config_set.py --role rover`), so the ESP32 drives the XBee's DIN through the header pin marked XBEE RX / GPS TX2 and forwards the corrections to the ZED over UART1. Build: `feather_s3_reversetft_socket`.

2. Power wiring



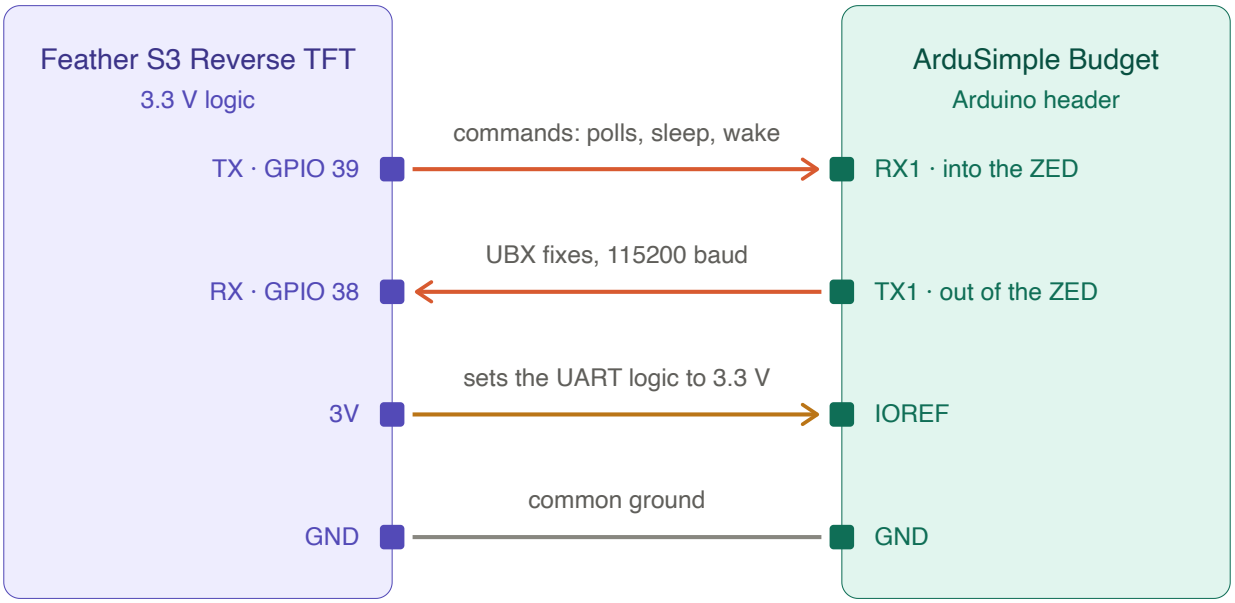
ArduSimple switch **down** (IOREF=N/C / 5V=INPUT): 5 V comes from the regulator and IOREF from the Feather's 3V. The XBee is powered by the socket. Leave the S9V11F5's EN unconnected; its pull-up keeps it on. Remove the 10 kΩ test resistor on the RX2 pin.

3. Feather pins



The Feather seen from the header side, USB-C at the top. GPIO 9 (SLEEP_RQ) is unused with the XBee in the socket. GPIO 11 to the regulator's EN is optional, only for cutting the GPS and XBee power in deep sleep (`-DGPS_POWER_PIN=11`).

4. GPS link detail



Each side's TX goes into the other side's RX

ArduSimple names TX1 and RX1 from the ZED's side: TX1 is the ZED's output and goes to the Feather's RX.

5. Connection list and setup

From	To	Notes
LiPo 10,000 mAh	Feather JST socket	check the + marking first
Feather BAT (right edge, top)	S9V11F5 VIN	3.0–4.2 V
S9V11F5 VOUT	ArduSimple 5V IN	switch down
S9V11F5 EN	nothing (or GPIO 11)	optional deep-sleep power cut
Feather 3V (left edge, 2nd)	ArduSimple IOREF, LIS3DH VIN	3.3 V logic
GND (left edge, 4th)	S9V11F5, ArduSimple, LIS3DH	all common
ArduSimple TX1	Feather RX (GPIO 38)	UBX at 115200
Feather TX (GPIO 39)	ArduSimple RX1	commands and forwarded RTCM
XBEE TX / GPS RX2 pin	Feather GPIO 5	XBee DOUT: what the base sends
Feather GPIO 6	XBEE RX / GPS TX2 pin	XBee DIN: telemetry out
LIS3DH SDA / SCL	Feather SDA / SCL	address 0x18
LIS3DH INT	Feather GPIO 10	motion wake

Set up beforehand, from a computer

- **ZED:** `python3 ubx_config_set.py --port <ZED port> --role rover` (UART1 UBX at 115200, UART2 off, saved to flash). A Pi rover uses `--role rover-pi` instead.
- **XBee:** BD = 115200, SM = 0 (no SLEEP_RQ line in the socket).
- **Firmware:** `pio run -e feather_s3_reversetft_socket -t upload`